

AI/ML Project - Smart School ECAM

FKY

2026

Introduction

Congratulations! You have been selected by the school management for a new digitalization project! The project called **Smart School** wants to use data mining and ML techniques on the school internal data to reach the goal of the organization, supporting our students in the acquisition of new competences. The Smart School project is divided into two parts:

1. Failure prediction: The school wants to identify students who are likely to fail an exam so they can provide them with special tutoring and support. To do this, they collected a dataset containing information about students and past exams, and they ask you to build an ML model to predict student grades. School management will then review students with a predicted grade below 50 and provide additional support.
2. Automatic correction: Grading is time-consuming and repetitive for teachers, so they want a system to automate correction of exams and tests. They are interested in using OCR (Optical Character Recognition) to recognize students' handwritten responses and correct them automatically. A training dataset of characters and digits is provided for this task.

The different datasets are available in the project folder on Claco. Additional information is given for the Advanced AI Labs evaluation at the end of this assignment.

Practicalities

The project will be evaluated based on a report (**maximum 10 pages**) and the code produced by the student. It will be completed with an **oral defense** after the theoretical oral exam in June. The report should be well-structured with sections corresponding to the steps of your methodology. Each section should clearly describe what was done, include graphs, and discuss the results critically. **Groups of two are recommended**, but individual work is also acceptable. Maintain a critical analysis throughout the project and discuss limitations, rather than a purely descriptive approach. The submission deadline on Claco is on the **6th of May**. **An evaluation grid is provided below; read and use it**. During the oral evaluation you do not need a formal presentation, but you will need to answer questions about your methodology, report contents, and code.

Tips

Here are a few tips before getting started:

- **READ THE EVALUATION GRID.** It describes the expected structure and lists several requirements.
- EDA requires multiple graphs to be complete, you can put some in annex.

- When evaluating your failure prediction model, use a baseline (for example, the naive average of the students' grades). This allows you to compare your model with a very simple "worst-case" model and determine whether your approach performs better.
- For evaluating the OCR system, consider using a confusion matrix.

Evaluation Grid

Legend:

- **Absent** – No evidence of the criterion.
- **Insufficient** – Attempt made but poorly executed or misunderstood.
- **Sufficient** – Meets the basic expectations.
- **Good** – Above average, solid execution.
- **Excellent** – Outstanding, insightful, and well-justified work.

Criteria	Absent	Insufficient	Sufficient	Good	Excellent
Failure prediction: Data Exploration and Analysis (15%)					
Understand dataset structure	Not done	Minimal structure identified	General understanding	Detailed exploration with insights	Deep and thorough understanding
Exploratory Data Analysis (EDA)	Not done	Basic stats only	Basic EDA and visuals	Meaningful insights	In-depth EDA with patterns and trends
Data visualization	Not done	Poor or unclear visuals	Basic graphs	Relevant graphs with analysis	High-quality and insightful visuals
Feature importance analysis	Not done	Poor or unclear analysis	Basic importance discussed	Importance supported with logic or metrics	Advanced techniques well justified
Failure prediction: Feature Selection and Creation (5%)					
Relevant feature selection	Not done	Arbitrary or irrelevant features	Some useful features selected	Justified selection using data or domain logic	Optimal features chosen via analysis
Feature transformation / creation (Optional depending on the necessity)	Not done	Irrelevant or redundant features	Basic transformations or new features	Useful engineered features	Innovative and insightful feature creation
Failure prediction: Data Pre-processing (10%)					

Missing values / outliers	Not done	Poor or inappropriate handling	Reasonable handling	Good justification	Robust and nuanced approach
Categorical encoding	Not done	Inconsistent or flawed encoding	Basic encoding used	Proper encoding with logic	Advanced or balanced encoding strategies
Scaling numerical features	Not done	Incorrect or partial scaling	Appropriate scaling used	Good choice of method	Method adapted to model needs
Failure prediction: Model Comparison and Hyperparameter Selection (25%)					
Dataset splitting	Not done	Incorrect or biased splitting	Basic split (train/test)	Valid train/val/test split	Valid train/val/test split and cross-validation
Model training	No models trained	One or irrelevant model	2 models tested	3 models tested (among which one deep learning model)	more than 3 models tested (among which one deep learning model) with choice justification.
Performance comparison	Not done	Poor metrics or unclear comparison	Basic metrics compared	Structured and meaningful comparison	In-depth multi-metric evaluation
Hyperparameter selection/tuning	Not done	Poor or basic tuning	Basic cross-validation	Good tuning and validation	Advanced tuning with cross-validation
Over/underfitting analysis	Not done	Incorrect or shallow	Basic discussion	Correct analysis with remedies	Deep, justified understanding
Automatic Correction / OCR (25%)					
OCR dataset preparation	Not done	Poor or incomplete preparation	Basic preparation	Good preprocessing with logic	Complete and clean dataset ready for OCR
OCR model choice and training	Not done	Poor or irrelevant choice	Acceptable algorithm proposed	Good choice justified with known performance	Excellent choice with strong justification

Presentation of Results (20%)					
Report structure and visuals	Unreadable	Poorly structured or weak visuals	Basic structure and graphs	Clear structure, good visuals	Excellent clarity and presentation
Methodology and discussion	No method described	Weak or incoherent logic	Acceptable structure and explanation	Coherent and logical methodology	Rigorous and critically discussed methods
Performance evaluation	Not done	Vague or missing metrics	Basic performance metrics	Good metrics and interpretation	Excellent analysis with insights and comparison

Advanced AI Labs Supplement

For students following the 4eiai40 course with the Advanced AI labs, we ask you to extend your analysis of the project by taking into consideration the different aspects covered during the labs:

- **Security:** Some of the data used for failure prediction is provided by the students themselves. Could a dishonest student lie about certain variables in order to obtain additional tutoring, even if they are not among the most in need? Try to put yourself in the place of such a student and determine how one could manipulate the inputs while changing as little information as possible. Is there a risk for the OCR part?
- **Bias:** The dataset contains several variables. If sensitive variables are used, how do they affect the model? Do they introduce unwanted bias? If so, you should evaluate this bias and apply techniques to mitigate it while trying to maintain the best possible performance.
- **Explainability:** In the interest of transparency, a school may wish to explain to a student the factors that made them eligible for special support. We ask you to provide an explanation based on which and how much input variables influenced the decision. At the same time, and from a more scientific perspective, a teacher interested in the OCR model would like to know which parts of the image are used to predict the character. Could you provide such an explanation?

To respond to these aspects you have to submit a 5 page report supplement for the **21th of May**. **The attendance at the advanced labs is mandatory**, any unjustified absence will penalize your grade.